Computational cost and numerical feasibility of smoothness-indexed penalized least squares: an archived multi-solver study

Heriberto Espino-Montelongo[a], Daniela Cortés-Toto[a]

[a] *Universidad de las Américas Puebla, San Andrés Cholula, Puebla, Mexico*

ARTICLE INFO

Keywords:
Penalized least squares
normalized smoothness
computational benchmarking
selected inverse
numerical stability
solver selection
reproducible science

ABSTRACT

Finite-difference penalized least-squares (PLS) trends may be indexed by normalized smoothness $S \in [0,1]$ rather than a scale-dependent penalty λ. Selecting an efficient implementation requires more than one linear solve: the complete CV, GCV, AICc, BIC and marginal-likelihood workloads include grid inversion, factorization, exact trace or leverage diagnostics and setup reuse. We compare ten existing complete-criterion pipeline policies, including explicit dense inversion, full spectral SVD, banded Cholesky and banded, sparse or iterative hybrids. The completed, corrected-source Windows runs archive 5570 successful raw timing observations across 450 method–case attempts; 1102 additional workload records failed or were partial, 18 disagreed with a reference criterion score, and 72 lacked a valid spectral comparator. For the matched $n = 256, d = 1$, GCV workload at fixed $\mu = 0$, banded Cholesky took 8.13 ms versus 13.80 ms for SVD with one smoothness candidate, whereas spectral SVD took 26.68 ms versus 61.31 ms for Cholesky with 100 candidates (two timing repetitions on a single Windows runner). The same descriptive winner reversal appeared under the empirical-difference plug-in mean and fixed $\mu = 0.15$. An MLE-normalization discrepancy discovered in the initial pilot was corrected analytically, reducing recorded score disagreements from 104 to 16. We distinguish fitted-trend accuracy from criterion-score parity, preserve unavailable and unmeasured cases without fictitious times, and give a conditional fastest-feasible-method rule as an *unvalidated research hypothesis*, not a deployed dispatcher or an exhaustive factorial conclusion. All source revisions and raw outputs remain archived.

1. Introduction and smoothness coordinate

Quadratic finite-difference penalized least squares (PLS) has a long history in Whittaker–Henderson smoothing (Whittaker, 1923; Guerrero, 2007). The penalty λ is mathematically convenient, but the same numerical value need not express the same smoothness for different difference orders or sample lengths. Guerrero's trace-based smoothing measure (Guerrero, 2008) and its use with CV, GCV, AICc and BIC (Cortés-Toto, Guerrero and Reyes, 2017) are established precedents. We do not claim to invent the smoother or its tuning criteria; instead, we investigate the *computational cost of an interpretable bounded coordinate and the reuse of its mapping across many evaluations.*

For $y \in \mathbb{R}^n$, difference order $1 \le d < n$, and fixed mean of the penalized differences μ, let $D_d \in \mathbb{R}^{(n-d)\times n}$ denote forward differences and define

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau}\{\|y-\tau\|_2^2 + \lambda\|D_d\tau - \mu\mathbf{1}\|_2^2\}. \quad (1)$$

Put $r = n-d$, $Q_d = D_d^\top D_d$, and $H_\lambda = (I+\lambda Q_d)^{-1}$. A polynomial particular solution is $p_{\mu,t} = \mu\binom{t-1}{d}$, and $\widehat{\tau} = p_\mu + H_\lambda(y - p_\mu)$. The penalty has a d-dimensional polynomial nullspace, so $\mathrm{tr}(H_\lambda) \to d$ as $\lambda \to \infty$.

ORCID(S):

Write $\delta_1, \dots, \delta_r > 0$ for the positive eigenvalues of Q_d. We use the normalized smoothness index

$$S_d(\lambda) = \frac{n - \mathrm{tr}(H_\lambda)}{n-d} = \frac{1}{r}\sum_{j=1}^{r}\frac{\lambda\delta_j}{1+\lambda\delta_j}. \quad (2)$$

Each $\delta_j > 0$ implies $S_d'(\lambda) = r^{-1}\sum_j \delta_j/(1+\lambda\delta_j)^2 > 0$, hence S_d is continuous, strictly increasing, and maps $[0,\infty]$ onto $[0,1]$. Every interior s has a unique penalty $\Lambda_{n,d}(s)$. The map depends only on (n,d), *not* on observations or the fixed reference mean μ. Therefore it can be cached across different series, criteria and μ values. The denominator $n-d$ normalizes an established trace-based index; it is not a new statistical model.

1.1. Shift-free full spectral reconstruction

Let $D_d = U\,\mathrm{diag}(\sigma)V^\top$ be a reduced SVD over the r positive singular modes and write $w_j = \lambda\sigma_j^2/(1+\lambda\sigma_j^2)$. A numerically useful identity is

$$\widehat{\tau}_{\lambda,\mu} = y - V\big[w \odot (V^\top y) - \mu(w \oslash \sigma) \odot (U^\top\mathbf{1})\big], \quad (3)$$

where multiplication and division are componentwise. It avoids forming the potentially enormous polynomial p_μ at high d and costs $O(nr)$ for a complete curve once the SVD is available. A separate high-precision companion manuscript evaluates its numerical accuracy. Here we study *runtime costs* without treating a numerically invalid fast method as a winner.

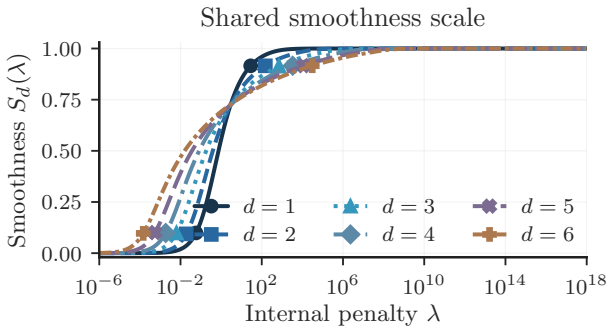


Figure 1: Smoothness versus penalty for $n = 48$, orders $d = 1, \ldots, 6$. The sixth-order line illustrates the mathematical mapping, not a completed Windows timing benchmark for $d = 6$.

The exact endpoints are interpolation at $s = 0$ and the polynomial-constrained limit at $s = 1$, which is evaluated analytically rather than factorizing a matrix at $\lambda = \infty$. Figure 1 illustrates why a user benefits from a common unit-interval smoothness coordinate even though the corresponding penalty scales vary with d.

1.2. Higher-order differences and physical interpretation

When y_t represents a sampled position at equally spaced times $t_j = j\Delta t$, the rescaled finite differences $(D_d\tau)_j/(\Delta t)^d$ approximate the dth time derivative as $\Delta t \to 0$ for sufficiently smooth trajectories. Thus $d = 1$ penalizes departures from constant velocity (first differences), $d = 2$ from constant acceleration (second differences), and $d = 3$ from constant jerk (third differences). This interpretation requires that the measured variable is position and the sampling step is meaningful; for generic economic or environmental series these are merely algebraic difference orders.

At the maximum-smoothness limit, the penalty constraint is $D_d\tau = \mu\mathbf{1}$. For $\mu = 0$, the limiting fit belongs to the polynomial nullspace of D_d and has degree at most $d - 1$: $d = 1$ gives a constant, $d = 2$ a straight line, $d = 3$ a quadratic and $d = 4$ a cubic. For a *nonzero fixed* μ, the limiting fit has degree d (for example, $d = 3$ permits a cubic with constant third difference, corresponding to constant nonzero jerk for uniformly spaced positions). In the physical units of position, $\mu/(\Delta t)^3$ approximates jerk; μ alone is not a jerk value without units and spacing.

1.3. Research questions

A single fitted curve, one criterion score, a complete grid search, and R independent searches are different workloads. We distinguish: setup costs; full fitted-trend reconstruction; trace and leverage evaluation for CV/GCV/AICc/BIC/MLE; one-point evaluation; matched 17-point grids; locally optimized continuous searches; and repeated-series reuse. Spectral preprocessing may be expensive for one query yet efficient when amortized. The question is not whether s magically lowers the cost of optimizing λ, but whether the *same exact set* $\Lambda_{n,d}(s_j)$ can be calculated once, reused, and exposed to users as interpretable $s_j \in [0, 1]$.

2. Algorithmic time and storage complexity

Let n be series length, d difference order, $r = n - d$, K penalty candidates, R observed series, C scoring criteria and I root-solver evaluations per requested s. All big-O statements are arithmetic models, not measured wall-clock rates. The penalty matrix $Q_d = D_d^\top D_d$ has half-bandwidth d.

2.1. Factorization and one complete fitted curve

Full SVD of the $r \times n$ difference matrix requires $O(nr^2)$ time and $O(nr)$ storage, or $O(n^3)$ and $O(n^2)$ respectively when d is small relative to n. Dense symmetric eigendecomposition of Q_d also has cubic time and quadratic storage; forming Q_d can exacerbate spectral conditioning. After spectral setup a full fitted curve uses $O(nr)$ products. Dense general solves and dense Cholesky require $O(n^3)$ per new penalty. General banded solving and banded Cholesky can factor in $O(nd^2)$ and use $O(nd)$ storage, followed by $O(nd)$ triangular or back substitution. Thus structured methods can be substantially faster for a *single fit*; Cholesky and general banded LU must not be conflated.

2.2. Criterion evaluation is not the same as fitting

For a fixed λ, the conditional effective degrees of freedom are $\nu(\lambda) = d + \sum_j (1 + \lambda\delta_j)^{-1}$ and normalized smoothness is (2), both computed in $O(r)$ after spectral preparation. A projected observation $z = V^\top(y - p_\mu)$ costs $O(nr)$ once, after which $\mathrm{RSS}(\lambda) = \sum_j w_j(\lambda)^2 z_j^2$ and the conditional GCV, AICc, BIC and profiled Gaussian marginal likelihood require $O(r)$ spectral sums per candidate. Classical fixed-μ LOOCV also requires leverages $(H_\lambda)_{ii}$ and residuals: calculating both using stored right singular vectors costs $O(nr)$ per penalty. This is *conditional observation-weight deletion*, not rolling-origin forecast validation, and empirical plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$ would not have the same exact leave-one-out interpretation without refitting that mean.

Nonspectral methods may fit quickly while incurring additional exact EDF and leverage work. Simply solving n banded right-hand sides to obtain inverse diagonals costs up to $O(n^2 d)$ beyond the $O(nd^2)$ factorization; specialized selected-inverse or structured algorithms can improve this (Weinert, 2007; Biessy, 2026). We distinguish the actual timed implementations from these more specialized competitors and

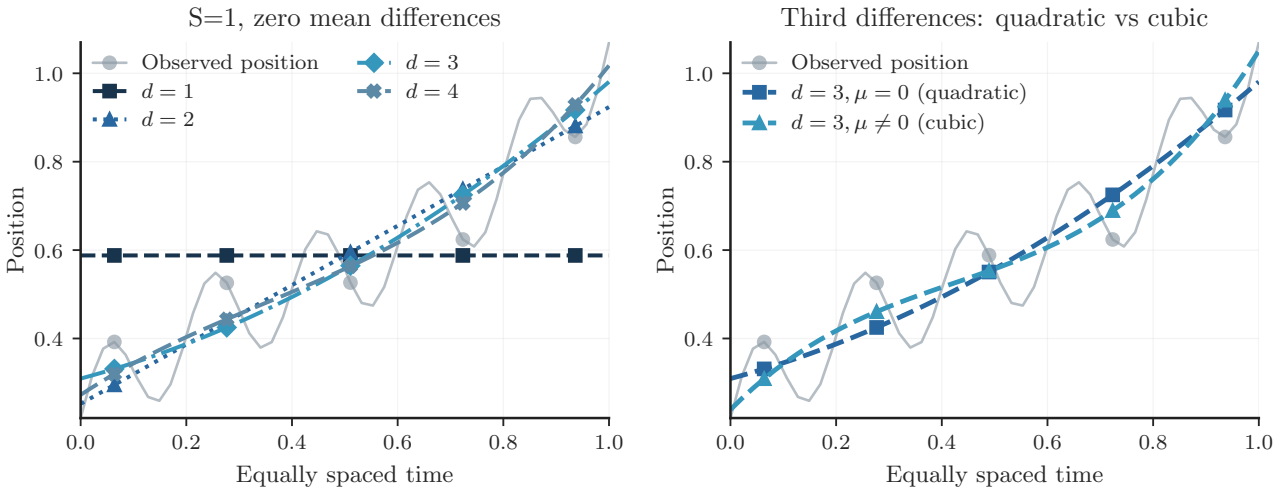


Figure 2: Deterministic illustration of limiting ($S = 1$) polynomial trends on a sample position trajectory, not a timing experiment. Left: zero-mean dth differences give polynomial degree at most $d - 1$. Right: $d = 3$, $\mu = 0$ gives a quadratic; a nonzero fixed third-difference mean gives a cubic, which corresponds to constant jerk only after scaling by the time step cubed.

Table 1
Dominant arithmetic for the implemented algorithm families. Costs are conditional on fixed μ and known finite λ. EDF and CV entries for nonspectral methods are not interchangeable with their fit cost. We state $r = n - d$ and assume a dense spectral basis is retained.

Method	Decomposition/setup	One fit	EDF/GCV	LOOCV	Memory
Fresh dense solve	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
Cached dense solve	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
Dense Cholesky	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
General banded LU	$O(nd^2)$	$O(nd^2)$	method-specific	method-specific	$O(nd)$
Banded Cholesky	$O(nd^2)$	$O(nd^2)$	method-specific	method-specific	$O(nd)$
Dense eigendecomposition	$O(n^3)$	$O(n^2)$	$O(n)$	$O(n^2)$	$O(n^2)$
Full SVD of D_d	$O(nr^2)$	$O(nr)$	$O(r)$	$O(nr)$	$O(nr)$

avoid advertising a fit-only cost as an end-to-end CV-selection cost.

For dense direct methods, EDF and LOOCV may be computed using a factorization and many right-hand sides; the displayed cubic bound reflects this full operation. A properly implemented banded selected inverse is not necessarily governed by the straightforward many-right-hand-side cost. These costs exclude software-level allocation and numerical conditioning, both measured or audited separately.

2.3. How pretranslation and grid reuse change total work

With the singular values cached, one inverse $s \mapsto \lambda$ is a monotone scalar root problem in $\log \lambda$ whose function calls each take $O(r)$, so K targets with I iterations cost $O(KIr)$ beyond the SVD. Let the set $G_{n,d} = \{\Lambda_{n,d}(s_j) : 1 \leq j \leq K\}$ be cached. It depends neither on the current series nor on the fixed mean of differences. The same grid can be reused across C criteria and R new series.

For GCV/AICc/BIC/ML-style spectral selections, an illustrative amortized cost is

$$T_{\text{SVD}} = O(nr^2 + KIr + Rnr + RCKr). \tag{4}$$

The Rnr term projects each new series once; CV replaces the pertinent $O(r)$ candidate term by $O(nr)$. A cold approach that re-inverts s at every criterion or re-decomposes for each series incurs additional repeated work. Likewise, a banded Cholesky implementation that caches the same K factors across R series may incur $O(Knd^2 + RKnd)$ fitting arithmetic, *before* trace/leverage computation. This is why different setup/reuse policies can reverse a timing winner.

The reference-mean policy changes coefficients of the right-hand side, not the spectrum. Fixed $\mu = 0$ and fixed $\mu = 0.15$ share precisely the same $G_{n,d}$. Estimating $\widehat{\mu}$ from $D_d y$ adds an $O(nd)$ pass per series; the archived comprehensive Windows run did *not* separately time this data-dependent mean-estimation step.

2.4. What is actually maximized under the Gaussian ML criterion?

The benchmark's label *ML* means profiled Gaussian *marginal* variance-component likelihood, not pointwise maximum likelihood of the fitted curve. It assumes independent observational errors and a Gaussian penalty prior,

$$\begin{aligned} y &= \tau + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2 I), \\ D_d\tau - \mu\mathbf{1} &\sim N(0, \sigma_\eta^2 I), \quad \lambda = \sigma_\varepsilon^2/\sigma_\eta^2. \end{aligned} \tag{5}$$

The polynomial nullspace is unpenalized and eliminated from the reduced likelihood. On the $r = n - d$ penalized singular coordinates z_j, the marginal variances are $\sigma_\varepsilon^2 c_j(\lambda)$ with $c_j = 1 + 1/(\lambda\delta_j)$. Profiling the unknown observation variance gives

$$\begin{aligned} \widehat{\sigma}_\varepsilon^2(\lambda) &= \frac{1}{r}\sum_{j=1}^{r} z_j^2/c_j(\lambda), \\ -2\ell_{\mathrm{prof}}(\lambda) &= r\log\widehat{\sigma}_\varepsilon^2(\lambda) + \sum_{j=1}^{r}\log c_j(\lambda) + \text{constant}. \end{aligned} \tag{6}$$

This is a REML-like reduced-coordinate likelihood and it *does* depend on a variance model. After selection, one may also compute $\widehat{\sigma}_\eta^2 = \widehat{\sigma}_\varepsilon^2/\widehat{\lambda}$. The benchmark uses fixed $\mu = 0$ or 0.15; any data-derived $\mu = \mathrm{mean}(D_d y)$ would instead constitute a conditional plug-in, *not* joint maximum likelihood over $(\mu, \lambda, \sigma_\varepsilon^2, \sigma_\eta^2)$. For fixed variances and λ, maximizing the Gaussian posterior in τ recovers the ordinary PLS trend; that is a *different* optimization from selecting λ by marginal ML.

2.5. Matched versus unmatched coordinate searches

We separate *native s grid* (invert each s_j during a search), *premapped s grid* (invert once, pass the identical λ_j to the score), and an *independently spaced log-λ grid*. Only the first two evaluate the same statistical candidates, and hence they must agree in exact arithmetic. The third may run faster but selects among *different interior penalties*; its search time alone cannot establish an efficiency advantage at equal statistical quality. Continuous bounded optimizers in s and $\log\lambda$ follow different paths and are local searches, not certified global optima. Exact $s = 0$ and $s = 1$ endpoints are scored analytically.

The classical residual-based BIC expression has a divergent interpolation boundary; our numerical companion studies this limitation. Here BIC timings are purely computational and do not imply the unrestricted criterion is statistically well-posed.

3. Structured end-to-end criterion computation

A fast linear-system solution is not automatically a fast smoothing-parameter selector. For each candidate penalty, an implementation must obtain the trend, its residual sum of squares and, depending on the objective, the effective degrees of freedom (EDF), the diagonal leverage values or the marginal likelihood determinant. These auxiliary calculations are included in our definition of a *complete computational pipeline*. We give a structured construction that avoids a dense $n \times n$ inverse.

3.1. Banded factorization and the selected inverse

Let $A_\lambda = I + \lambda Q_d$, $Q_d = D_d^\top D_d$, and $\lambda \in (0, \infty)$. The symmetric positive definite matrix A_λ has half-bandwidth d. Factor it as $A_\lambda = L_\lambda L_\lambda^\top$ with lower triangular L_λ, whose nonzero entries satisfy $0 \le k - i \le d$. Define $Z_\lambda = A_\lambda^{-1}$. Although Z_λ is generally dense, only the entries Z_{ij} with $|i - j| \le d$ are needed in the selected-inverse recurrence. Backward substitution from $i = n$ to 1 gives

$$Z_{ij} = -\frac{1}{L_{ii}}\sum_{k=i+1}^{\min(n,i+d)} L_{ki} Z_{kj}, \quad i < j \le \min(n, i+d), \tag{7}$$

and

$$Z_{ii} = \frac{1}{L_{ii}^2} - \frac{1}{L_{ii}}\sum_{k=i+1}^{\min(n,i+d)} L_{ki} Z_{ki}. \tag{8}$$

Symmetry supplies Z_{kj} when $k > j$. These are selected-inverse (Takahashi-type) identities (Erisman and Tinney, 1975), not a proposal to store the full inverse. A banded factorization and selected inversion each require $O(nd^2)$ arithmetic, with $O(nd)$ factor and selected-entry storage for a fixed small difference order. The leverage diagonal is $h_i = Z_{ii}$ and $\nu(\lambda) = \sum_i h_i$.

The finite-penalty right-hand side is constructed directly:

$$b_{\lambda,\mu} = y + \lambda\mu D_d^\top \mathbf{1}, \qquad A_\lambda \widehat{\tau} = b_{\lambda,\mu}. \tag{9}$$

This bypasses the potentially huge polynomial particular solution p_μ, avoiding one source of catastrophic cancellation at large d and nonzero μ. It does not, however, guarantee forward accuracy when A_λ becomes badly conditioned. We require finite outputs, residual checks and independent reference comparisons before accepting a timed result.

3.2. Conditional criteria and computational requirements

For a *fixed numeric* μ define $e = y - \widehat{\tau}$, $\mathrm{RSS} = e^\top e$ and $h_i = (H_\lambda)_{ii}$, with $H_\lambda = A_\lambda^{-1}$. The classical leave-one-observation-out criterion and generalized cross-validation are

$$\mathrm{CV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1-h_i}\right)^2, \qquad \mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}/n}{(1-\nu(\lambda)/n)^2}. \tag{10}$$

The original residual-based criteria used in the accompanying benchmark have the form

$$\begin{aligned} \mathrm{BIC}_{\mathrm{RSS}}(\lambda) &= \log(\mathrm{RSS}/n) + \nu(\lambda)\log n/n, \\ \mathrm{AICc}(\lambda) &= \log(\mathrm{RSS}/n) + \frac{2\nu(\lambda)+1}{n-\nu(\lambda)-2}. \end{aligned} \tag{11}$$

We retain the exact score conventions of the existing software rather than silently substituting a different information criterion. The BIC expression decreases without bound as $\lambda \downarrow 0$ under a nondegenerate interpolation limit; the finite grid lower boundary must therefore be disclosed. The profiled Gaussian marginal-likelihood criterion additionally requires a weighted quadratic form and a log determinant on the penalized subspace, described in Section 2. Its implementation is not interchangeable with merely calculating a residual MSE.

If μ is estimated from the same observations as $\widehat{\mu} = \mathrm{mean}(D_d y)$, then H_λ is no longer the complete response-to-fit derivative of the plug-in procedure. The standard fixed-μ leverage formula in (10) is *not* an exact leave-one-out refit of this estimated mean. Our planned fixed-mean comparisons distinguish $\mu = 0$ and $\mu = 0.15$; plug-in μ is a separately labeled conditional marginal-likelihood analysis.

3.3. Sparse and iterative methods with honest diagnostic costs

We benchmark nine end-to-end *pipelines*, not nine interchangeable Cholesky routines. Sparse SuperLU, general banded LU, SuiteSparse CHOLMOD, and preconditioned conjugate-gradient (PCG) each provide a fitted trend through a different algorithm. In the current research code their CV/GCV/AICc/BIC/ML scores are completed by an *additional* banded selected-inverse diagnostic step. Thus the hybrid runtime includes both the named fit backend and the added diagnostic factorization. SciPy sparse LU is not CHOLMOD; an unavailable optional CHOLMOD installation is recorded as such.

PCG uses a matrix-free application of $I + \lambda D_d^\top D_d$ and FFT-periodic, DST-Dirichlet or Jacobi preconditioning. The FFT and DST symbols approximate the natural finite-series boundary structure; they are not exact diagonalizations, and no fixed small iteration count is guaranteed. Nonconverged PCG runs cannot enter the successful-time comparison.

4. Experiment006: factorial evaluation protocol

The present expansion treats *complete selection time* as its principal response. Statistical estimation error, including trend RMSE, is outside this experiment. Numerical validity remains an eligibility condition: a faster output is not a successful computation if it fails the stated mathematical equations. We distinguish the *prospective* Experiment006 design from archived small-n observations reported in Section 6. This section describes a partially implemented research protocol; the large factorial has not been measured.

4.1. Four crossed workloads

Write M for the number of candidate smoothness values, R for independent series sharing (n, d), and C for the number of criteria evaluated. Four workload classes cross the two relevant dimensions: The first workload exposes cold setup costs; the second exposes amortization across penalties; the third isolates reuse across distinct observed vectors; and the fourth measures a realistic multi-series selection batch. In particular, a grid-level speedup is not extrapolated from timing one isolated factorization.

4.2. Factors, endpoints and numerical scales

Because $S_d(0) = 0$ and $S_d(+\infty) = 1$, the endpoint $s = 1$ is a *constrained* PLS limit, not a very large finite penalty. Any unstable high-order polynomial projection at this endpoint is recorded as a numerical failure. At $n = 131072$ a single dense float64 $n \times n$ array requires 128 GiB before factors or workspaces; dense inversion and full SVD must be explicitly marked resource-ineligible at that scale, not dropped from the design.

Small and large observed amplitudes alone do not define the numerical problem: fixing a nonzero μ while rescaling y changes the relative strength of the imposed difference-mean constraint. The factorial therefore crosses amplitude and μ rather than treating their scales as exchangeable. With $d = 6$ the nonzero-μ limiting polynomial can have very large coefficients even when the data values are small.

4.3. Fair coordinate comparison

Define a common candidate set $\mathcal{G} = \{s_1, \ldots, s_M\}$ and its image $\Lambda_{n,d}(\mathcal{G})$. In the on-demand implementation, s_j is inverted whenever requested. In the pretranslated implementation all λ_j are computed once and passed directly to the same backend criterion. Therefore,

$$F_{\mathrm{direct}}(s_j) = F_{\mathrm{translated}}\big(\Lambda_{n,d}(s_j)\big) \tag{12}$$

in exact arithmetic for any one criterion F and fixed reference mean. Identical selected values (within numerical tolerance) are a *correctness check*; relative time

Table 2
Leading arithmetic complexity for one finite-penalty, fixed-μ complete criterion evaluation with difference half-bandwidth d and $r = n - d$. For hybrid implementations, additional banded diagnostics are *charged*, even if their fit operation is inexpensive. q is the attained PCG iteration count; these are arithmetic orders, not measured timings.

Complete pipeline	Preparation / factor	Criterion	Memory
Fresh dense inverse	$O(n^3)$	$O(n^2)$ after inverse	$O(n^2)$
Full SVD (after one decomposition)	$O(nr^2)$ once	$O(r)$ for GCV/AICc/BIC/ML; $O(nr)$ for CV	$O(nr)$
Banded Cholesky + selected inverse	$O(nd^2)$	$O(nd)$ for solve and criterion	$O(nd)$
Banded or sparse LU + banded trace	fit-backend dependent $+O(nd^2)$	$O(nd)$ after diagnostics	fit memory $+O(nd)$
CHOLMOD + banded trace	sparse-factor dependent $+O(nd^2)$	$O(nd)$ after diagnostics	sparse-factor memory $+O(nd)$
PCG + banded trace	$O(qnd)$, or $O(qn(d + \log n))$ with transforms, $+O(nd^2)$	$O(nd)$ after diagnostics	$O(nd + n)$, excluding FFT workspaces

Table 3
Four full-pipeline workloads. A single S is 0.5; the complete grid uses the levels in Table 4. A criterion score, not merely a fit, is timed.

Workload	M	R
Single value, single series	1	1
Grid, single series	$M > 1$	1
Single value, multiple series	1	$R > 1$
Grid, multiple series	$M > 1$	$R > 1$

quantifies reuse of conversion, not a change of statistical estimand. Independently spaced log-λ grids sample different models and require a separate sensitivity analysis.

Let T_0 be solver preparation, T_Λ the once-computed grid mapping, T_p the data projection for one series, and T_F one criterion score. For a spectral GCV-like objective a stylized cost model is

$$T_{\text{shared}} \approx T_0 + T_\Lambda + RT_p + RCMT_F, \tag{13}$$

whereas repeated on-demand inversion can introduce up to RCM mapping calls. These are accounting models, not measured speedups. CV changes T_F through leverages; sparse and hybrid algorithms incur separate factorization, cache and diagnostics costs. A fair comparison fixes (n, d, μ, R, M), criteria and candidates, then records preparation, mapping, per-series projections, candidate scoring, total time and retained memory.

4.4. Paired measurement and analysis

Each run records source revision, CPU, operating system, Python/SciPy/BLAS versions, thread counts, warm-up policy, candidate count, random seed and every attempted computation. Independent seeded timing blocks with counterbalanced method order are the proposed publication protocol. Only complete, numerically valid attempts enter winner comparisons. Planned summaries include means, medians, standard deviations, confidence intervals by independent timing block and paired speedups against explicit dense inversion wherever feasible. For fixed levels of other factors, a regression $\log T = \alpha + \beta \log n$ across at least three series lengths gives a descriptive empirical scaling exponent, not a proof of big-O complexity.

The runnable design manifest, bounded slices and mathematical tests reside in the separate Experiment006 research branch. The largest $M = 100000$ by $n = 131072$ combinations require resource gating, not extrapolation from completed pilots. Incomplete and ineligible cells are scientifically meaningful observations, not measured timing winners.

5. Archived benchmark design and provenance

This section analyzes *completed and committed historical* timing experiments; the prospective Experiment006 factorial in Section 4 has different axes and is not claimed as measured. It does not substitute fitted complexity curves for measured time or combine machine-specific durations from different hosts into a single speedup. The principal dataset was collected on the author's Windows 11 workstation (the environment identifies Windows build 26100) with Python 3.11.17, NumPy 2.4.6, SciPy 1.17.1, OpenBLAS limited to one thread, two experimental seeds and seven timing repetitions per reported configuration.

The complete Windows experiment contains 80,640 raw timing records and a separate computing-environment record; precise repository paths and checksums are

Table 4
Target axes of Experiment006. This is a *planned design*, not completed measurements. Each new smoothness grid includes $s = 1$ but starts above zero; historical 17-point grids also included $s = 0$.

Axis	Levels / interpretation
Series length n	$64, 128, 256, 512, 1024, 16384, 131072$
Difference order d	$1, 2, 3, 4, 5, 6$
Reference mean	Fixed $\mu = 0$ or $\mu = 0.15$; observed-difference plug-in $\widehat{\mu}$ separately, conditional ML only
Observation magnitude	Low (10^{-3} times a fixed seeded shape) and high (10^{6} times the same shape)
Criterion	Conditional CV, GCV, AICc, residual-based BIC, profiled Gaussian marginal ML
Smoothness grid	$\{j/M : \ j = 1, \ldots, M\}$, $M = 100, 1000, 10000, 100000$, including $s = 1$ exactly
Selection coordinate	On-demand $s \mapsto \lambda$ inversion; once-pretranslated $\lambda_j = \Lambda_{n,d}(s_j)$ and identical candidate scoring
Complete pipeline	Naive inverse, SVD, banded Cholesky/selected inverse, and six named LU/CHOLMOD/PCG plus banded-diagnostic hybrids
Reuse	$R = 1$ or $R > 1$ independent series; independent paired timing blocks, cold/warm accounting and cache-memory constraints

Table 5
Actual coverage of the archived timing studies. The principal full-factorial rows are *not* 80,640 independent trials: they comprise 40,320 design cells and two recorded seeds. Additional repetitions are timed measurement blocks. This table separates which extra data would still be required to claim all requested combinations.

Corpus	Measured design	Limitation
Full Windows search	$n = 24, 48$; $d = 0, \ldots, 5$; $\mu = 0, 0.15$ fixed; $R = 1, 3, 8$; CV/GCV/AICc/BIC/ML; seven backends; eight search policies; cold/shared setup; 2 seeds; 7 repetitions	No $d = 6$ timing or empirical $\widehat{\mu} = \mathrm{mean}(D_d y)$
20-seed solver study	SVD, dense eigendecomposition, banded Cholesky; $M = 1, 16, 64$ full curves; $n = 32, 96, 192$ plus selected $d = 1, 3, 5$ cells; 20 independent seeds	Fit-only, not five-criterion selection
GCV reuse study	Six solver families; uniform-s versus log-λ grid; fixed means 0 and 0.15; varied grid sizes and series reuse	GCV only; generally different grid interiors
Extended fit/trace studies	Dense fresh/cached, spectral, general banded and Cholesky alternatives; selected n up to 400, M up to 256	Earlier machine/candidate regimes; not pooled with Windows factorial

listed in the accompanying `SOURCE_AUDIT.md`. A checked analysis yields 40,320 two-seed design cells and verifies exactly 4,484 matched native-s/premapped-grid valid seed pairs with *zero difference* in selected smoothness and score. The associated `windows_full_s_analysis/` folder preserves checks and per-cell paired costs. We use only cells whose complete evaluated grid passed the backend's recorded numerical validation; an implementation that returned a partial grid does not win a speed ranking.

5.1. Solver families and workloads

The seven Windows backends are: fresh dense solve, cached-Q dense solve, general banded solve, dense Cholesky, banded Cholesky, eigendecomposition of Q_d and full SVD of D_d. The five criteria are classical fixed-μ CV, GCV, AICc, residual-based BIC and the implemented conditional marginal-likelihood score. Eight policies cover one point at $s = 0.5$ with native/premapped/direct penalties; complete 17-point native-s, premapped and independent log-λ grids; and bounded *local* optimization in s and $\log \lambda$. All policies are evaluated at $R \in \{1, 3, 8\}$ and cold/shared preparation. Grid conversion is separately timed; a standalone premapped search charges it once, while reuse across criteria charges it only once across the entire joint workflow.

The full normalized-s grid includes the exact $s = 0$ and $s = 1$ endpoints plus near-one finite points through $s = 1 - 10^{-12}$. For large d these correspond to potentially extreme finite penalties; failed factorizations and numerically inaccurate criterion values are explicitly tracked. The Gaussian likelihood used in this corpus was marked *not applicable* for $d = 0$. Although that historical run includes order zero, all smoothness theory and the proposed new methodological scope in this manuscript use $1 \leq d < n$.

For the representative timing table, the original script selected $n = 48, d = 2, \mu = 0, R = 1$, SVD, and the median over its two Windows seeds. The log-λ grid has 17 points but different interior values, and the continuous policies are not guaranteed global optimizers. No inferential result follows from comparing differently discretized search spaces.

Table 6
Five completed 17-point selection searches for $d = 3$, $n = 48$, fixed $\mu = 0$, SVD, $R = 1$. Values are two-seed medians in milliseconds. Both policies evaluate the identical full $S \in [0, 1]$ grid; premapped standalone time includes conversion once per criterion.

Criterion	Repeated S inversion	Premapped S grid
CV	2.058	1.846
GCV	1.958	1.723
AICc	2.024	1.724
Residual BIC	1.972	1.724
Profiled Gaussian ML	2.038	1.776

5.2. Statistical interpretation of runtime measurements

A two-seed Windows timing median describes that host and work unit; it is *not* a sample of 80,640 independent processors or experiments. The separate solver study used 20 independent data seeds and paired timings to compute pointwise confidence intervals and Holm-corrected sign-flip tests for prespecified fitted-trend workloads. We report these uncertainty summaries separately rather than apply the study's p-values to the different Windows factorial. Setup, grid preparation and criterion scoring have different accounting boundaries, so their elapsed times must not be summed or divided without matching definitions.

The archived sources also do not contain an end-to-end, all-criteria $d = 6$ timing factorial, nor an additional $\widehat{\mu} = \mathrm{mean}(D_d y)$ timing policy. These are concrete extensions for the final combined paper, not silently filled gaps. The measured numerical-error companion covers $d = 6$ but cannot be converted into a runtime observation after the fact.

6. Measured computational results

6.1. One full S-grid, with or without reusable inverse mapping

The principal case now uses $n = 48, d = 3, \mu = 0, R = 1$, the SVD backend and five selection criteria. Both policies evaluate *exactly the same 17 normalized-smoothness candidates*, including $S = 0$ and $S = 1$: native-S repeatedly invokes $\Lambda_{n,d}$, whereas the cached policy translates the grid once and reuses its penalties. This is an equal-estimator computational comparison, not a contest between different statistical grids. The recorded two-seed median timing values are inserted automatically from the original Windows CSV, not manually estimated. Although the candidate grid includes $S = 0$ and $S = 1$, some criterion values at those endpoints can be mathematically undefined (notably residual BIC at interpolation), and those scores are not reinterpreted as admissible optima.

In the jointly reused five-criterion $d = 3$ workflow, the native-S total was 10.050 ms, versus 3.767 ms for one shared mapping plus five backend-only searches (ratio 2.668). In contrast, the premapped column of Table 6 charges the mapping once *per standalone criterion*; summing those values would overcount shared preparation.

For a second, historically reported $d = 2$ comparison at the same $n = 48, \mu = 0, R = 1$, the archived five native-S searches total 11.234 ms, versus 3.980 ms when the grid is mapped once and shared across five backend-only searches: a descriptive factor of 2.823. These $d = 2$ values are *not* relabeled as $d = 3$ timings; the new $d = 3$ table and plot use their own archived records. Within each difference order, a precomputed grid can be shared across criteria, series and fixed μ values without altering the PLS estimator.

6.2. One evaluation, full grid and repeated independent series

Figure 4 plots all seven backends at a *single* $s = 0.5$ GCV score, one mapped 17-point GCV grid and eight independent full-grid GCV searches, under fixed $n = 48, d = 3, \mu = 0$. It uses the original full-factorial Windows rows, drawing a timing only where both experimental seeds completed that particular workload with valid numerical scores. Unsuccessful or incomplete cases are missing, not assigned fictitious times. The figure makes visible why a one-evaluation ranking need not generalize to selection or reuse workloads: decomposition cost is paid once, while data projections, curve solves and per-candidate trace work scale differently across algorithms.

6.3. Repeated grid reuse and difference order

Figure 5 shows the recorded ratio of cold to shared backend-only time for $R = 8$ complete s-grid searches across $d = 1, \ldots, 5$, conditional $\mu = 0$, $n = 48$, and the SVD backend. Differences among criteria are attributable to distinct scoring operations and reuse boundaries, not a change in statistical PLS formulation. A ratio greater than one favors shared setup for that cell; outcomes missing the necessary two valid seeds are left blank, not silently imputed.

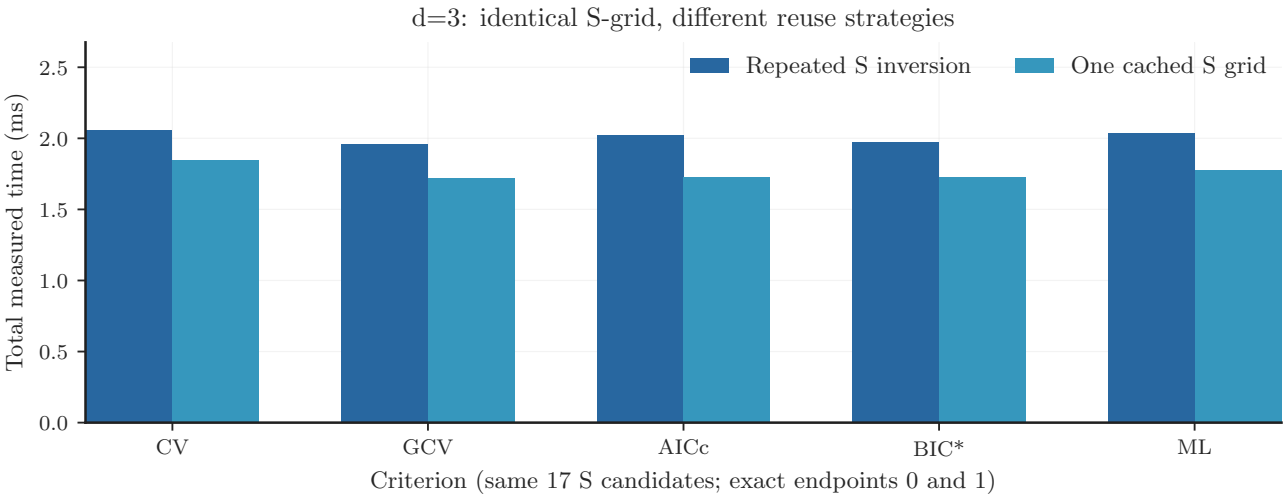


Figure 3: Full $S \in [0,1]$ selection at $d = 3$, $n = 48$, fixed $\mu = 0$ and SVD, with five criteria: repeated on-demand $S \mapsto \lambda$ inversion versus one cached candidate translation. Both policies evaluate identical penalties. Bars are observed two-seed median standalone timing results, including one-time mapping cost in the second policy.

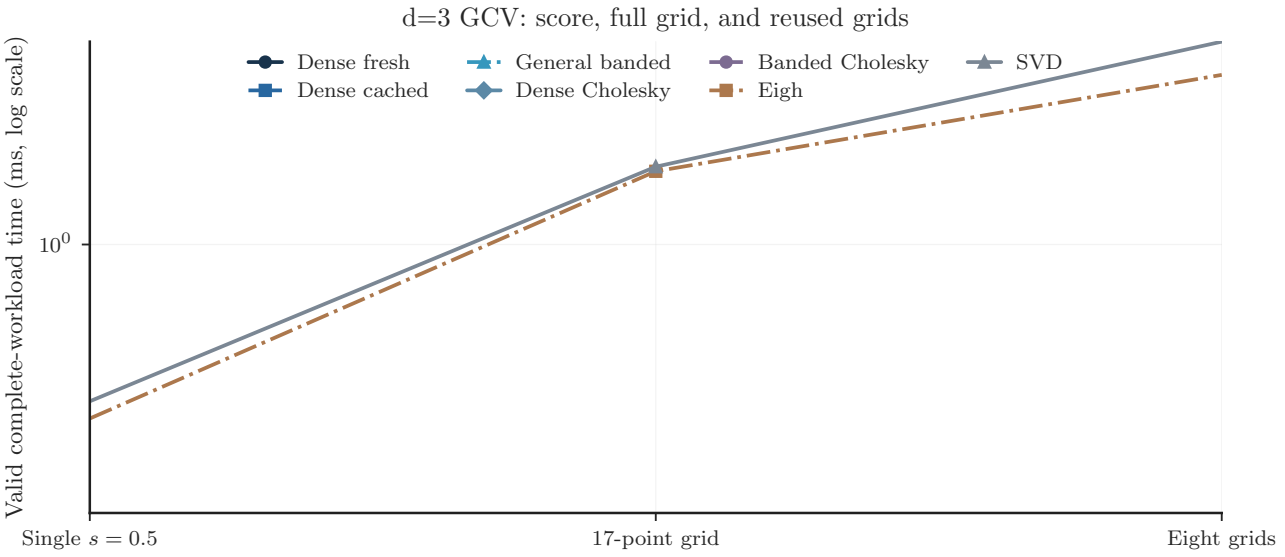


Figure 4: Observed standalone elapsed time for one GCV score ($s = 0.5$), one 17-point grid, and eight independent grids, across seven implementations. Data are from the two-seed Windows full factorial with $n = 48, d = 3, \mu = 0$ and shared preparation. All seven methods supplied a valid single-score evaluation, but only two supplied valid complete full-grid and repeated-grid cases at this difficult $d = 3$ setting. Points are shown only for workloads with complete numerically valid seed pairs; missing points are not zero elapsed time or a performance win. The three workload classes are distinct algorithms, not repeated scalar multiplications of one measured time.

6.4. Longer series and repeated fitted curves

The separate complete-trend benchmark, with n extending to 400 and up to 256 repeated fits, documents substantial changes in winners. For $n = 400, d = 2, \mu = 0$, the ratio of fresh dense to spectral total time was 0.158 at $M = 1$ but 28.314 at $M = 256$; 0.158 means the dense approach was faster for the single fit, whereas 28.314 means the spectral implementation was faster than the repeatedly fresh dense solver on the larger sweep. Against a properly general banded solver the corresponding ratios were 0.055 and 0.339, so *banded remained faster* in those two recorded workloads. These are host-specific paired cases, not universal crossovers and not five-criterion selections.

The independent 20-seed solver study also supports comparing higher orders. Figure 6 shows the original paired geometric-mean SVD/banded-Cholesky total-runtime ratio at $n = 96$, $M = 16$ complete fitted curves, fixed $\mu = 0$, and orders $d \in \{1, 2, 3, 5\}$. There is *no*

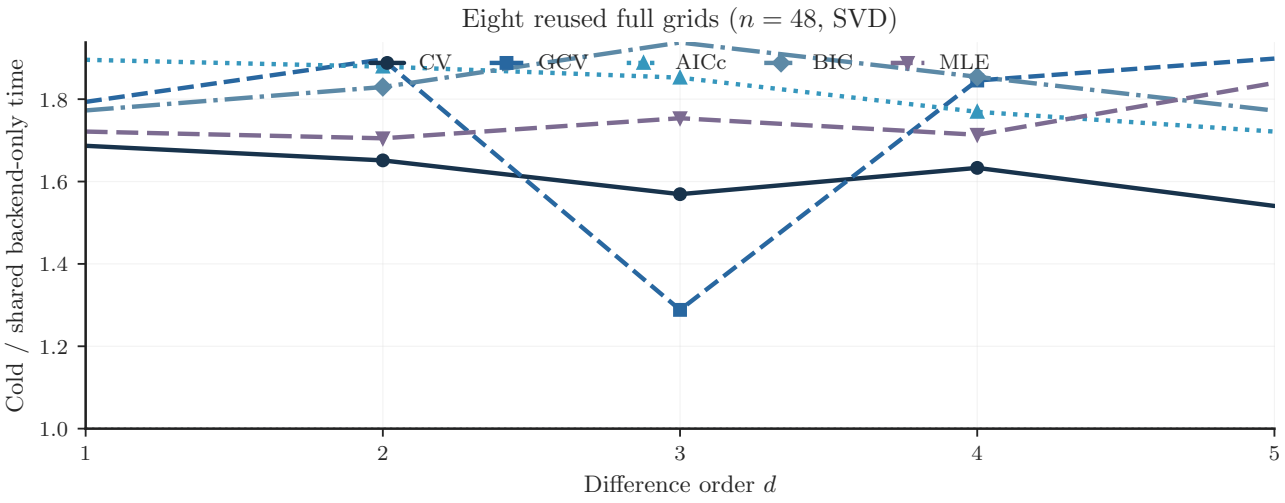


Figure 5: Cold versus shared implementation cost for eight repeated matched-grid selections by difference order and criterion. A vertical value above one indicates less backend-only time with shared preparation. $n = 48$, fixed $\mu = 0$, SVD; exact source is the archived two-seed Windows timing CSV. No $d = 6$ Windows timing was recorded.

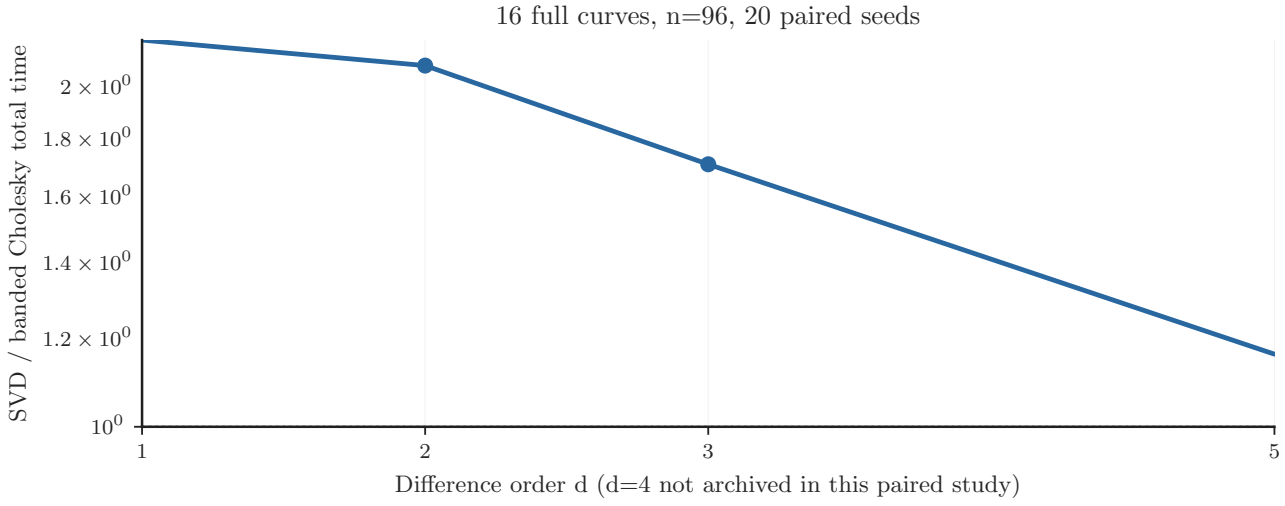


Figure 6: Measured paired SVD/banded-Cholesky full-curve total-time ratios across available difference orders at $n = 96$, $M = 16$ and $\mu = 0$. Each order has 20 independent paired data seeds. $d = 4$ was not included in this matched solver workload; no result is extrapolated. This is not a criterion-selection benchmark.

paired $d = 4$ observation for this exact workload in that source: the figure does not interpolate or fabricate one. A ratio above one favors banded Cholesky and below one SVD. These are *fit-only* timings and do not cover trace, leverage or five-criterion selection.

6.5. Exact matched-grid equivalence

The complete source audit verified 4,484 numerically valid seed-matched native-s/premapped-λ grid comparisons with *zero* differences in selected s and scored values. This is expected from (2): a monotone coordinate transform does not change the statistical model. The computational advantage lies in *the caching policy*, not statistical changes. Results from unmatched log-λ or bounded continuous search cannot establish this equivalence because they evaluate different candidate sets.

No measured cell in this section should be considered an objective benchmark at $d = 6$ or for data-derived $\widehat{\mu}$; those remain future research dimensions.

7. Earlier evidence status and planned comparisons

This section preserves the original experimental hypotheses and evidence snapshot from Experiment006. The subsequently completed, bounded Experiment010 measurements are analyzed in Section 8, not silently

Table 7
Evidence classification as of this manuscript version. A green software test or successful small pilot does not establish the full-factorial timings. In particular, the only historical joint five-criterion timings here are from $n = 48$.

Question	Available evidence	Remaining evidence needed
Matched S versus translated-λ grids	4,484 valid matched two-seed cases in historical Windows source; equal selected candidates and scores	All requested M, n, d, fixed-μ and amplitude regimes
Five-criterion reuse across the same mapped grid	Historical $n = 48, d = 2, 3$, fixed $\mu = 0$, SVD cases	Repeated criteria and series at large n including nonzero μ
Single fit versus repeated fits	Historical full-fit sweeps and separate 20-seed solver study	Direct full-criterion comparisons with the same candidate grids and hardware
Sparse, banded and iterative complete criteria	Implemented reference pipelines and mathematical parity tests in Experiment006 research code	Repeated, paired end-to-end timings and numerical validation on PCAG103A-08
Maximum smoothness $S = 1$ and $d = 6$	Analytic limit and selected-inverse verification for small n	High-order independent accuracy checks at larger n and high data magnitude
Asymptotic scaling vs measured exponents	Derived arithmetic/storage bounds	Three or more measured n levels per comparable factor cell, confidence intervals and clear resource-ineligible statuses

pooled with the historical corpus. The archived Windows timing corpus in Section 6 covers $n = 24, 48$ and $d \leq 5$, whereas Experiment006 expands to seven series lengths and all six positive difference orders. The two data sources must not be pooled into one purported factorial cell or used to construct an unmeasured empirical complexity exponent.

Our first hypothesis is that a banded Cholesky plus selected-inverse pipeline will avoid the quadratic memory and cubic setup costs of full dense spectral or inverse-based approaches when d is fixed. The second is that full SVD will become competitive in configurations with enough repeated candidate scores to amortize its initial decomposition. The third is that one-time translation of a fixed S grid will reduce mapping overhead across independently observed series, with identical selected S values for complete, valid matched grids. The fourth is that higher orders, strong smoothness and nonzero reference means can change the numerical failure pattern enough that a raw elapsed-time ranking becomes misleading without an accuracy gate. These are *testable research hypotheses*, not conclusions extrapolated to unmeasured factor levels.

For the full factorial, the primary observed response is end-to-end time in seconds, conditional on numerical correctness. Secondary responses are setup time, one-time mapping time, score time, peak memory, number of root solves, number of PCG iterations and the fraction of ineligible or failed attempts. When the naive dense baseline cannot run, a speedup against that baseline is *undefined*; it must never be replaced with a synthetic time predicted from $O(n^3)$. Mean and confidence intervals are reported across independent paired blocks rather than treating tightly correlated inner-loop stopwatch samples as independent observations.

Table 8
Independent high-order fitted-trend accuracy audit ($n = 128$, fixed $\mu = 0.15$). A method rejected by its own convergence check is not counted as a valid fit. These are accuracy observations, not speed rankings.

Fit backend	Valid/28	Rejected/28	Largest rel. error
Shift-free full SVD	28	0	6.18×10^{-9}
Banded Cholesky	16	12 inaccurate	21.24
Sparse LU	22	6 inaccurate	6.86×10^{-2}
FFT PCG	14	14 nonconverged	7.64×10^{-12}

7.1. Independent high-order numerical audit (completed pilot)

A distinct, immutable Experiment006 audit on the Windows self-hosted `PCAG103A-08` runner compared fixed-μ PLS fits with an independent 85-decimal-digit dual banded-Cholesky reference, including a 55-versus-85-digit precision-convergence check. The source and all 112 backend-level observations are preserved in GitHub branch `results/experiment006-highd-41439239921`, Actions run `41439239921`. Its factor levels were $n = 128$, $d \in \{5, 6\}$, $\mu = 0.15$, two independent frozen seeds, and $s \in \{0.75, 0.85, 0.90, 0.91, 0.95, 0.99, 1\}$. The numerical validity threshold here is relative *forward error* at most 10^{-6} against the independent high-precision fit, not an arbitrary double-precision normal-equation residual.

The audit demonstrates why a small double-precision KKT residual does *not* establish a small forward error at high difference order and high smoothness. Consequently, Experiment006 now exposes an explicitly named `guarded_svd_cholesky` benchmark policy: shift-free full SVD for $d \geq 5$, and banded Cholesky for $d \leq 4$. Its SVD setup and memory costs are fully included; dense resource guards make it ineligible at

very large n. This policy is empirically supported in the present restricted $n = 128$ audit, not a universal large-series stable solution.

The subsequent guarded Windows pilot, Actions run `41439589986`, preserved 80/80 valid observations in the $d = 2$, $\mu = 0$ GCV slice, and 56/80 valid observations in the $d = 6$, $\mu = 0.15$ CV slice. The 16 newly added guarded observations were valid, while the prior 24 failures from unguarded methods were retained. Neither this pilot nor the independent accuracy audit supplies the requested seven-length complete factorial or publication-grade speedup confidence intervals.

This distinction also guides manuscript revision. Future Experiment006 results will be inserted only after their raw timing CSV, environment metadata, numerical validity summary, and source revision have been committed to the repository. The large seven-length factorial remains incomplete, while the subsequent verified bounded multi-backend observations are now incorporated in Section 8. The present article therefore documents a completed experimental snapshot, not a completed exhaustive factorial.

8. Completed multi-backend experiments

This section reports only the experiments completed and archived by 3 November 2026. The planned larger factorial is not presented as measured, and the historical small-series results in Sections 5–6 are not pooled with the newer Windows timing observations.

8.1. Full-criterion workloads

The experiment varies series length n, finite-difference order d, candidate smoothness count M, number of independent series R, criterion C (CV, GCV, BIC, AICc or Gaussian marginal likelihood), reference mean μ, observation amplitude, native-S versus pretranslated-λ evaluation, and the complete numerical backend. Three mean policies are compared:

$$\mu = 0, \qquad \widehat{\mu}_{\mathrm{data}} = \frac{\mathbf{1}^\top D_d y}{n - d}, \qquad \mu = c = 0.15.$$

The sample-derived mean is computed separately for each observed series. It produces conditional plug-in scores, not exact leave-one-out re-estimation of the mean or joint marginal maximum likelihood. For the same (n, d), all three policies share the same exact map $\Lambda_{n,d}(S)$, independent of the observations and mean policy.

All complete-criterion pipelines charge their setup, smoothness mapping, fit, effective degrees of freedom, leverage and marginal-likelihood diagnostics as applicable. The guarded SVD/Cholesky policy is composite; sparse and PCG hybrid pipelines include their extra banded diagnostics. An incomplete, unavailable or numerically invalid method is never assigned a fictitious valid execution time.

8.2. Completed attempts and archival accounting

All new measurements used the existing non-administrator `trendestimation` Conda environment on university Windows 11 runner `PCAG103A-08`. The committed source SHA, original `raw_times.csv`, per-method criterion scores and failure statuses, seed, environment and batch summary are preserved in immutable GitHub result branches. A status-`ok` timing row is a method–workload–coordinate–repetition observation, *not* an independent experimental problem.

The pilot advanced only 300 out of 1800 scheduled method–case attempts. The 50,400-method–case and 403,200 workload/policy-row factorial *manifests* remain plans, not completed time observations. The original 80,640-row historical Windows study, Experiment007 fit-only runs and other Experiment008/009 measurements remain separate datasets.

8.3. All ten pipelines under a matched smoothness grid

Table 10 compares the corrected $n = 256, d = 1$ GCV experiment: $\mu = 0$, high input amplitude, one observed series, and pretranslated λ corresponding to *identical* S candidates for every numerical backend. The reported number is the median *total cold* time in milliseconds over two repetitions, including grid mapping, factorization and the extra criterion diagnostics where required.

Naive dense inversion completed the single-S evaluation but not the complete 100-point workload within the resource gate. Optional sparse CHOLMOD was not installed. The guarded policy uses banded Cholesky at $d = 1$ and should be regarded as a composite, not a separate factorization algorithm. The numerical pipelines therefore exhibit a *descriptive* reversal: banded Cholesky led at one S, whereas spectral SVD led at 100 candidates. This is not an established universal computational crossover.

The same pattern was recorded for all three mean policies: These two-repetition measurements describe one implementation and host and cannot support a population-level confidence interval. At $n = 256, d = 5$, some structured 100-candidate MLE pipelines did not finish with completely acceptable criterion records, whereas SVD and the guarded policy did. At $n = 512$, 72 timing records lacked a valid matched spectral reference. They must remain unverified rather than declared correct.

8.4. High-order reliability and a discovered MLE normalization error

The independent Experiment006 85-decimal-digit fitted-trend audit (Table 8) used $n = 128$, $d = 5, 6$, fixed $\mu = 0.15$, and 28 reference problems per backend. Under its 10^{-6} relative forward-error threshold, full shift-free SVD passed 28/28, banded Cholesky 16/28,

Table 9
Verified corrected-source Experiment010 evidence. Match means selected-score agreement against a valid spectral reference, not independent high-precision verification. Failed/partial records have no valid complete-criterion time. No-reference rows cannot be independently compared with spectral scores.

Batch	Attempts	OK times	Match	Differ	No ref.	Failed/partial
Corrected smoke	30	420	420	0	0	30
Initial corrected pilot	60	720	704	16	0	180
Pilot continuation offset 60	60	680	680	0	0	208
Pilot continuation offsets 120–240	180	2520	2520	0	0	180
$n = 256$ targeted frontier	60	666	664	2	0	222
$n = 512$ targeted frontier	60	564	492	0	72	282
Total	**450**	**5570**	**5480**	**18**	**72**	**1102**

Table 10
Ten complete-criterion policies in one matched Windows case ($n = 256, d = 1$, GCV, $\mu = 0$), measured median cold milliseconds. A dash means the optional method was unavailable or the full candidate workload did not produce a valid completed measurement; it does not represent an infinite or estimated time.

Pipeline	One S	100 S candidates
Naive inverse	9.74	—
Shift-free spectral SVD	13.80	**26.68**
Banded Cholesky + selected inverse	**8.13**	61.31
Guarded SVD/Cholesky composite	8.26	58.85
Banded LU + banded diagnostics	8.55	63.38
Sparse LU + banded diagnostics	9.71	72.89
Sparse CHOLMOD + banded diagnostics	—	—
FFT-preconditioned PCG + diagnostics	9.09	73.63
DST-preconditioned PCG + diagnostics	8.30	79.87
Jacobi-preconditioned PCG + diagnostics	8.39	147.84

Table 11
Measured cold median milliseconds, $n = 256, d = 1$, GCV; one series and two repetitions.

μ	Chol. 1	SVD 1	Chol. 100	SVD 100
0	8.13	13.80	61.31	26.68
data plug-in	8.4	13.5	60.2	25.6
0.15	8.1	12.9	59.5	24.9

sparse LU 22/28 and FFT–PCG 14/28 accepted its convergence gate. This is separate *fit accuracy* evidence, not a high-precision certificate for each Experiment010 criterion timing.

The original *pre-correction* Experiment010 pilot reported 104 disagreements in selected criterion scores; 100 occurred for $n = 64, d = 5$, data-derived μ, and the profiled Gaussian marginal-likelihood criterion. The original finite-penalty banded, hybrid and naive benchmark implementations omitted $\log\det(D_d D_d^\top)$ from that criterion, whereas the spectral score and exact $S = 1$ endpoint retained it. For $r = n - d$ and positive eigenvalues δ_j of $D_d^\top D_d$,

$$\log\det(I + \lambda D_d^\top D_d) - r\log\lambda = \sum_{j=1}^{r} \log\left(1 + \frac{1}{\lambda\delta_j}\right) + \log\det(D_d D_d^\top). \tag{14}$$

Although the last term is constant among *finite* penalty candidates, omitting it destroys objective-value comparability with the correctly normalized exact $S = 1$ candidate. The implemented correction evaluates this term without a dense determinant using the identity

$$\det(D_d D_d^\top) = \prod_{j=0}^{d-1} \frac{\binom{n+j}{2j+1}}{\binom{2j}{j}}. \tag{15}$$

After revising the MLE benchmarks and regression tests, the repeated pilot had only 16 score disagreements, all from the high-order naive inverse. The remaining discrepancies and forward-error risks are not dismissed by score parity. Old and corrected raw results are preserved on distinct immutable GitHub branches. Correcting a statistical criterion is a scientific finding, not a claimed speedup.

8.5. Conditional computational minimum: a testable hypothesis

For a future workload w, define the set of eligible complete pipelines

$$\mathcal{F}(w) = \{b : b \text{ installed, memory-feasible, criterion-complete, and independently accuracy-certified}\}. \tag{16}$$

An operational hypothesis is to estimate the machine-specific full execution time $\widehat{T}_b(w)$—including setup, $S \mapsto \lambda$ mapping, factorization, criterion diagnostics and reuse—then choose

$$\widehat{b}(w) \in \arg\min_{b \in \mathcal{F}(w)} \widehat{T}_b(w). \tag{17}$$

On failure, a future dispatcher would try the next valid, feasible candidate and record extra time. The existing results do *not* establish a calibrated ten-method model, independent class-balanced held-out prediction of crossovers, or measured fallback regret. Leading-order arithmetic alone cannot predict actual execution time without constants, conditioning, iterative-solver convergence and memory constraints. Thus (17) is a *research proposal*, not a deployed, empirically proven selector.

8.6. Reproducibility and precise limits

The corrected-source Actions runs are `41587467639` (smoke and initial pilot), `41587723422` (pilot continuations) and `41642508020` (targeted $n = 256, 512$ frontier). The pre-correction failure-discovery run is `41586066141`. Each individual result branch and precise case path is indexed by `paper/csda-computational-costs/OBSERVED_RESULTS_SNAPSHOT.md` and `paper/benchmarks/experiment010/RESULTS_INDEX.md`. The full original CSVs, including failures and no-reference statuses, are the data source rather than any plotted or theoretical extrapolation. No execution at $n = 16384$ or 131072 and no completion of a 10,000–100,000-point search is inferred from the smaller archived batches. Numerical certification, large-series feasibility and the general conditional winner rule remain separate research questions.

9. Discussion

The difference between a mathematical coordinate and a computational policy is fundamental. The monotone smoothness map $S \leftrightarrow \lambda$ preserves the PLS estimator at corresponding penalty values. It does not make an arbitrary optimization search faster. Reuse of a translated candidate grid can reduce repeated work, as illustrated by the historical matched-grid $n = 48$ five-criterion observations: a 2.668-fold reduction for $d = 3$ and 2.823-fold for $d = 2$. These are host-specific examples, not universal speedup guarantees.

The completed Experiment010 measurements extend this distinction to the choice of *complete numerical backend*. The $n = 256, d = 1$ GCV results in Table 10 demonstrate that a fast structured factorization may lead for one selected smoothness, whereas the reusable spectral decomposition can lead for a 100-candidate search. This was also observed across the three recorded reference-mean policies. The work requested—one fit, one score, a grid search, or grid reuse over independent series and criteria—is therefore an essential part of algorithm choice.

9.1. Why a complete pipeline matters

The ten named benchmark pipelines are not ten new algebraic factorizations. The guarded policy selects an existing primitive, and sparse/iterative hybrids include the cost of an additional banded selected-inverse diagnostic factorization. A fit-only PCG or sparse LU solve cannot be compared fairly against a spectral GCV, CV or marginal-likelihood search with different outputs. The structure-exploiting methods of Weinert (2007) and recent Whittaker–Henderson developments by Biessy (2026) are essential prior art. Our table describes the actual Python implementation, including its additional diagnostics and memory gates, rather than the best conceivable implementation of each numerical family.

9.2. Why numerical validity precedes elapsed time

High-order differencing can make the finite-penalty systems very ill-conditioned. The independent 85-digit fitted-trend audit, the incomplete high-order criterion grids, the remaining naive-inverse score disagreements, and the missing $n = 512$ spectral-reference comparisons show why a fast recorded time is not automatically an admissible solver-selection outcome. In particular, the MLE correction (14) documents a statistical-criterion inconsistency: omitting a constant from finite-penalty likelihoods can alter selection once the exact $S = 1$ limit is evaluated on a different scale. The old observations were not erased and the corrected observations were not backdated.

9.3. Limits of the completed empirical study

The completed corrected-source dataset consists of 450 method–case attempts, not all 50,400 planned method–case entries. The 5570 successful timing rows include repeated, related workload observations on one Windows runner and cannot be treated as independent statistical replications. The two-repeat $n = 256$ comparisons warrant descriptive ranking, not tight confidence intervals for the prevalence of a winner across hardware. The measured $n \leq 512$ targeted cases do not justify extrapolating a dense SVD victory to $n = 16384$ or 131072, where quadratic storage may make it infeasible. The CHOLMOD optional backend was unavailable, so its speed cannot be inferred from sparse

LU. The sample-derived reference mean is a conditional plug-in, not the true latent difference mean and not jointly estimated ML. The classical CV criterion is not chronological forecast cross-validation; this paper does not measure forecast RMSE. The residual BIC interpolation limit remains statistically problematic regardless of the numerical backend.

Equation (17) gives a precise decision-theoretic target: minimize predicted full execution time over the subset of independently certified, available, memory-feasible pipelines, with explicit fallback on runtime failure. However, the present archive does *not* validate the model constants, holdout crossover classification, or regret of such a dispatcher. Those are future experimental questions, not a hidden conclusion of the present study.

10. Conclusions

An interpretable normalized smoothness index can be used without changing the underlying PLS estimator; its computational benefit comes from exact mapping and reuse. The completed ten-policy archive identifies different measured winners for single-candidate and full-grid GCV at $n = 256, d = 1$ under three different reference-mean policies. It also documents concrete high-order failures and an analytically corrected marginal-likelihood normalization inconsistency. The resulting research contribution is a *reproducible, numerically gated, evidence-limited computational comparison*, not a universal fastest solver or a demonstrated automatic selector.

Data and code availability

Original historical timings, the tested algorithms and reproducible figure sources are preserved in https://github.com/heri-espino/trendestimation. The corrected Experiment010 source and result index are under `paper/benchmarks/experiment010/`, with immutable `results/experiment010-*` branches. The corrected Windows Actions runs are 41587467639, 41587723422 and 41642508020; the pre-correction run is 41586066141. The manuscript companion `OBSERVED_RESULTS_SNAPSHOT.md` lists exact branch identifiers, observed counts, raw-file paths and the numerical limitations. Every reported timing is traceable to an original source CSV, not an estimated or filled-in value.

Declarations

Coauthor approval is still required for the provisional name order, affiliations, funding, conflicts and contribution statements before submission. This is a completed research snapshot, not a submitted journal manuscript.

References

Biessy, G., 2026. Whittaker–henderson smoothing revisited: A modern statistical framework for practical use. ASTIN Bulletin 56, 1–31. doi:10.1017/asb.2025.10061.

Cortés-Toto, D., Guerrero, V.M., Reyes, H.J., 2017. Trend smoothness achieved by penalized least squares with the smoothing parameter chosen by optimality criteria. Communications in Statistics—Simulation and Computation 46, 1492–1507. doi:10.1080/03610918.2015.1005236.

Erisman, A.M., Tinney, W.F., 1975. On computing certain elements of the inverse of a sparse matrix. Communications of the ACM 18, 177–179. doi:10.1145/360680.360704.

Guerrero, V.M., 2007. Time series smoothing by penalized least squares. Statistics & Probability Letters 77, 1225–1234. doi:10.1016/j.spl.2007.03.006.

Guerrero, V.M., 2008. Estimating trends with percentage of smoothness chosen by the user. International Statistical Review 76, 187–202. doi:10.1111/j.1751-5823.2008.00047.x.

Weinert, H.L., 2007. Efficient computation for whittaker–henderson smoothing. Computational Statistics & Data Analysis 52, 959–974. doi:10.1016/j.csda.2006.11.038.

Whittaker, E.T., 1923. On a new method of graduation. Proceedings of the Edinburgh Mathematical Society 41, 63–75. doi:10.1017/S0013091500077853.